Alexander Von Moll

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Education

University of Cincinnati — Cincinnati, OH, USA
PhD in Electrical Engineering — *2019 — 2022*

- GPA: 4.0
- Research: *Skirmish-Level Tactics via Game Theoretic Analysis*
- Advisor: Prof. Zachariah Fuchs

Georgia Institute of Technology — Atlanta, GA, USA
MS in Aerospace Engineering — *2014 — 2015*

- GPA: 4.0
- Special Project: *Machine Learning Applications in Complex Control Systems*
- Advisor: Prof. Evangelos Theodorou

The Ohio State University — Columbus, OH, USA
BS in Aero/Astro Engineering — *2008 — 2012*

- GPA: 3.97
- Minor: *Computer Science*

Work Experience

Aerospace Engineer — 2017 — Present
Air Force Research Laboratory — *WPAFB, OH, USA*

Perform fundamental research as a member of the UAV Cooperative Control Team under the Autonomous Controls Branch of the Power and Control Division. Describe, define, and solve problems of interest to the Air Force in the areas of pursuit-evasion differential games and algorithms for persistent intelligence, surveillance, and reconnaissance.

Aerospace Engineer — 2012 — 2017
Air Force Research Laboratory — *WPAFB, OH, USA*

Served as Integrated Product Team member under several turbine engine technology programs. Established plans and roadmaps for turbine engine control technologies. Supported research programs in turbine engine control software, actuators, and sensors.

Software Development Intern — Jun 2011 — Aug 2011
Boeing — *St. Louis, MO, USA*

Developed support software and tools for the F-15SA flight control software group.

Awards

- 2024 · Air Force Research Laboratory Jack Blackhurst Innovation Award
- 2022 · Excellent Reviewer – AIAA Journal of Guidance, Control, and Dynamics
- 2022 · University of Cincinnati Department of Electrical Engineering & Computer Science Outstanding Doctoral Dissertation Award
- 2019 · Aerospace Control & Guidance Systems Committee Dave Ward Memorial Lecture Award
- 2014 · AFRL Turbine Engine Division Civilian of the Year
- 2014 · Department of Defense SMART Scholarship
- 2011 · Department of Defense SMART Scholarship

Editorial Experience and Professional Service

- Associate Editor for the 2026 American Control Conference
- Invited Session Organizer for the 2026 American Control Conference: Multi-Agent Control & Coordination (double session)
- Invited Session Organizer for the 2026 SciTech Forum: Multi-Agent Control & Coordination (double session)
- Invited Session Organizer for the 2025 SciTech Forum: Multi-Agent Control & Coordination (double session)
- Invited Session Organizer for the 2024 SciTech Forum: Multi-Agent Control & Coordination (double session)

Publications

Direct Contributions to IEEE Transactions on Aerospace & Electronic Systems

Two contributions have been made to IEEE TAES. The first addresses a two-attacker versus turret scenario in which cooperation among the attackers is required in order to achieve their goal of neutralizing the turret [1]. The second addresses the problem of rendezvous and subsequent tracking of a target moving on a fixed course [2].

[1] **A. Von Moll**, D. Shishika, Z. Fuchs, and M. Dorothy, "The Turret-Runner-Penetrator Differential Game with Role Selection," *IEEE Transactions on Aerospace & Electronic Systems*, vol. 58, no. 6, pp. 5687–5702, May 2022, doi: 10.1109/TAES.2022.3176599.

[2] I. Weintraub, **A. Von Moll**, E. Garcia, D. Casbeer, and M. Pachter, "Surveillance of a Faster Fixed-Course Target," *IEEE Transactions on Aerospace & Electronic Systems*, May 2023, doi: 10.1109/TAES.2023.3237129.

Relevance to IEEE Transactions on Aerospace & Electronic Systems

Guidance and Control Systems

This research contributes to the design and analysis of guidance and control systems, specifically focusing on optimal control, differential games, and tactical maneuvering.

- *Tactical and Optimal Evasion Against Pure Pursuit*:
  - Characterized the pure pursuit guidance law for finite capture radius [3] and provided guidelines for successful evasion of a faster evader
  - Optimal evasion from two pursuers employing pure pursuit [4]
- *Cooperative Defense*: Investigated circular target defense differential games [5], cooperative turret defense [6], and multi-pursuer border defense [7]
- *Threat-Aware Maneuvering*: Addressed navigation around weapon engagement zones via a variety of methods including optimal control [8], [9], stochastic optimal control [10], and geometric methods [11], [12]
- *Tactical Game Theory*: Solved several turret-based and pursuit-evasion differential games, providing optimal high-level guidance (i.e., heading) to the agents [13], [14], [15], [16]

These works directly apply to the following areas within GNC: missile applications including national or theater defense systems, autonomous guidance, and applications of optimization.

Autonomous Systems

This research directly supports the TAES focus on UAV autonomous flight control, swarms, and path planning for autonomous missions.

- *Swarm & Fleet Coordination*: Developed geometric approaches for multi-pursuer single-evader games [17], [18]
- *UAV Interaction & Roles*: Analyzed the turret-runner-penetrator differential game with a focus on role selection for coordinated autonomous platforms [1]
- *Autonomous Path Planning*: Developed a genetic algorithm approach for UAV persistent visitation and surveillance of a sequence of targets [19]

Intelligent Systems

Many of the previously referenced works directly address the collaborative teaming aspect of the Intelligent Systems area (e.g., [1], [6], [7]). Additionally, there are several other works which address the aspect of operating aerospace systems in uncertain environments, particularly in the area of avoiding weapon engagement zones in the presence of wind [20], [21]. Recent work on optimal uncertainty quantification has shown very promising capabilities for the certification of aerospace systems with limited information [22].

Journal Papers

[23] T. Chapman, **A. Von Moll**, and I. E. Weintraub, "Safe Navigation in the Presence of Range-Limited Pursuers," *Control Systems Letters*, May 2026, doi: 10.1109/LCSYS.2025.3645221.

[24] P. Surve, S. D. Bopardikar, **A. Von Moll**, I. Weintraub, and D. Casbeer, "Heterogeneous Pursuit of an Active Target Under Sensing Constraints," *Control Systems Letters*, May 2026, doi: 10.1109/LCSYS.2025.3645033.

[6] **A. Von Moll**, D. Maity, M. Pachter, D. Shishika, and M. Dorothy, "Target Defense Using a Turret and Mobile Defender Team," *Control Systems Letters*, May 2026, doi: 10.1109/LCSYS.2025.3645833.

[25] G. Das *et al.*, "Dynamic Assignment Switching and Singular Surfaces in Multi-Agent Pursuit-Evasion Games," *Transactions on Automatic Control*, May 2026.

[26] D. Milutinović, **A. Von Moll**, S. G. Manyam, D. W. Casbeer, I. E. Weintraub, and M. Pachter, "Cooperative Pursuit-Evasion Games with a Flat Sphere Condition," *Open Journal of Control Systems*, Oct. 2025, doi: 10.1109/OJCSYS.2025.3604640.

[27] P. Surve, S. Bopardikar, **A. Von Moll**, I. Weintraub, and D. W. Casbeer, "Mutual Support by Sensor-Attacker Team for a Passive Target," *Open Journal of Control Systems*, Oct. 2025, doi: 10.1109/OJCSYS.2025.3612246.

[28] D. Shishika, **A. Von Moll**, D. Maity, and M. Dorothy, "Deception in Turret Defense Game: Information Limiting Strategy to Induce Dilemma," *Open Journal of Control Systems*, Sept. 2025, doi: 10.1109/OJCSYS.2025.3611726.

[21] D. Milutinović, **A. Von Moll**, I. E. Weintraub, and D. Casbeer, "Stochastic Optimal Avoidance of Multiple Engagement Zones," *Journal of Aerospace Information Systems*, May 2025, doi: 10.2514/1.I011593.

[29] C. Bakker, A. Rupe, **A. Von Moll**, and A. Gerlach, "Operator-Theoretic Methods for Differential Games," *Journal of Computational Physics*, Apr. 2025.

[30] I. Weintraub, **A. Von Moll**, D. Casbeer, and S. G. Manyam, "Virtual Target Selection for a Multiple-Pursuer--Multiple-Evader Scenario," *Journal of Aerospace Information Systems*, Jan. 2025, doi: 10.2514/1.I011510.

[12] **A. Von Moll** and I. Weintraub, "Basic Engagement Zones," *Journal of Aerospace Information Systems*, vol. 21, no. 10, pp. 885–891, Sept. 2024, doi: 10.2514/1.I011394.

[31] B. Jha, S. D. Bopardikar, **A. Von Moll**, and D. Casbeer, "Energy-Optimal Joint Motion Planning of a Pursuer-Turret Assembly," *Journal of Guidance, Dynamics, and Control*, Aug. 2024, doi: 10.2514/1.G008067.

[16] **A. Von Moll** and M. Pachter, "Complete Solution of the Lady in the Lake Scenario," *Dynamic Games and Applications*, July 2024, doi: 10.1007/s13235-024-00614-2.

[32] S. Bajaj, B. Jha, S. D. Bopardikar, **A. Von Moll**, and D. Casbeer, "Shortest Trajectory of a Dubins Vehicle with a Controllable Laser," *Transactions on Automatic Control*, Mar. 2024, [Online]. Available: https://arxiv.org/abs/2403.12346

[33] M. Dorothy, D. Maity, D. Shishika, and **A. Von Moll**, "One Apollonius Circle is Enough for Many Pursuit-Evasion Games," *Automatica*, Jan. 2024, doi: 10.1016/j.automatica.2024.111587.

[34] S. Bajaj, S. Bopardikar, E. Torng, **A. Von Moll**, and D. W. Casbeer, "Multi-vehicle Perimeter Defense in Conical Environments," *Transactions on Robotics and Automation*, Jan. 2024, doi: 10.1109/TRO.2024.3351556.

[13] **A. Von Moll**, Z. Fuchs, D. Shishika, D. Maity, M. Dorothy, and M. Pachter, "Turret Escape Differential Game," *Journal of Dynamics and Games*, vol. 11, no. 2, pp. 100–114, Sept. 2023, doi: 10.3934/jdg.2023012.

[35] P. M. Dillon, M. D. Zollars, I. E. Weintraub, and **A. Von Moll**, "Optimal Trajectories for Aircraft Avoidance of Multiple Weapon Engagement Zones," *Journal of Aerospace Information Systems*, vol. 20, no. 8, pp. 520–525, June 2023, doi: 10.2514/1.I011224.

[2] I. Weintraub, **A. Von Moll**, E. Garcia, D. Casbeer, and M. Pachter, "Surveillance of a Faster Fixed-Course Target," *IEEE Transactions on Aerospace & Electronic Systems*, May 2023, doi: 10.1109/TAES.2023.3237129.

[36] S. Bajaj, S. D. Bopardikar, **A. Von Moll**, E. Torng, and D. W. Casbeer, "Competitive Perimeter Defense with a Turret and Mobile Vehicle," *Frontiers in Control Engineering*, Feb. 2023, doi: 10.3389/fcteg.2023.1128597.

[3] **A. Von Moll**, M. Pachter, and Z. Fuchs, "Pure Pursuit with an Effector," *Dynamic Games and Applications*, p. 961, Nov. 2022, doi: 10.1007/s13235-022-00481-9.

[5] **A. Von Moll**, M. Pachter, D. Shishika, and Z. Fuchs, "Circular Target Defense Differential Games," *Transactions on Automatic Control*, vol. 68, no. 7, pp. 4065–4078, Oct. 2022, doi: 10.1109/TAC.2022.3203357.

[37] **A. Von Moll** and Z. Fuchs, "A Lock-Evade, Engage or Retreat Game," *None*, Oct. 2022.

[38] S. G. Manyam, D. W. Casbeer, **A. Von Moll**, and Z. Fuchs, "Shortest Dubins Paths to Intercept a Target Moving on a Circle," *Journal of Guidance, Control, and Dynamics*, June 2022, doi: 10.2514/1.G005748.

[1] **A. Von Moll**, D. Shishika, Z. Fuchs, and M. Dorothy, “The Turret-Runner-Penetrator Differential Game with Role Selection,” *IEEE Transactions on Aerospace & Electronic Systems*, vol. 58, no. 6, pp. 5687–5702, May 2022, doi: 10.1109/TAES.2022.3176599.

[39] D. Milutinović, D. W. Casbeer, **A. Von Moll**, M. Pacther, and E. Garcia, “Rate of Loss Characterization that Resolves the Dilemma of the Wall Pursuit Game Solution,” *Transactions on Automatic Control*, vol. 68, no. 1, pp. 242–256, Dec. 2021, doi: 10.1109/TAC.2021.3137786.

[40] I. E. Weintraub, **A. Von Moll**, E. Garcia, and M. Pachter, “Maximum Observation of a Target by a Slower Observer in Three Dimensions,” *Journal of Guidance, Control, and Navigation*, vol. 44, no. 3, Dec. 2021, doi: 10.2514/1.G005619.

[41] S. G. Manyam, D. W. Casbeer, **A. Von Moll**, and Z. Fuchs, “Curvature Constrained Paths to Intercept a Target Moving on a Circle,” *Transactions on Automation Science and Engineering*, Dec. 2020.

[42] E. Garcia, D. Casbeer, **A. Von Moll**, and M. Pachter, “Multiple Pursuer Multiple Evader Differential Games,” *Transactions on Automatic Control*, May 2020, doi: 10.1109/TAC.2020.3003840.

[43] J. L. Salmon, L. C. Willey, D. Casbeer, E. Garcia, and **A. Von Moll**, “Single Pursuer Multiple-Cooperative Evaders in the Border Defense Differential Game,” *Journal of Aerospace Information Systems*, Mar. 2020, doi: 10.2514/1.I010766.

[7] **A. Von Moll**, E. Garcia, D. Casbeer, S. Manickam, and S. C. Swar, “Multiple Pursuer Single Evader Border Defense Differential Game,” *Journal of Aerospace Information Systems*, Feb. 2020, doi: 10.2514/1.I010740.

[18] M. Pachter, **A. Von Moll**, E. Garcia, D. Casbeer, and D. Milutinović, “Cooperative Pursuit by Multiple Pursuers of a Single Evader,” *Journal of Aerospace Information Systems*, Feb. 2020, doi: 10.2514/1.I010739.

[44] M. Pachter, **A. Von Moll**, E. Garcia, D. Casbeer, and D. Milutinović, “Two-on-One Pursuit,” *Journal of Guidance, Control, and Dynamics*, vol. 42, no. 7, July 2019, doi: 10.2514/1.G004068.

[14] **A. Von Moll**, M. Pachter, E. Garcia, D. Casbeer, and D. Milutinović, “Robust Policies for a Multiple Pursuer Single Evader Differential Game,” *Dynamic Games and Applications*, no. 10, pp. 202–221, May 2019, doi: 10.1007/s13235-019-00313-3.

[17] **A. Von Moll**, D. Casbeer, E. Garcia, D. Milutinović, and M. Pachter, “The Multi-Pursuer Single-Evader Game: A Geometric Approach,” *Journal of Intelligent and Robotic Systems*, vol. 96, no. 2, pp. 193–207, Jan. 2019, doi: 10.1007/s10846-018-0963-9.

Conference Papers

[45] S. Shah, **A. Von Moll**, and I. Weintraub, “Strategic Evasion Using a Multi-Hypothesis Control Barrier Function,” in *Aviation Forum*, AIAA, June 2026.

[46] A. Nimmagadda, J. Park, **A. Von Moll**, I. Weintraub, and D. Casbeer, “Information-Theoretic Multi-Agent Path Planning for Risk-Aware Navigation Under Environmental Uncertainty,” in *Aviation Forum*, AIAA, June 2026.

[11] **A. Von Moll** and I. Weintraub, “Reactive Vehicle Guidance using Dynamic Maneuvering Cue,” in *AIAA SciTech*, AIAA, Jan. 2026. doi: 10.2514/6.2026-1199.

[22] A. Subramanian *et al.*, “Certification of Dynamical Systems using End-to-End Distributionally-Robust Uncertainty Quantification,” in *AIAA SciTech 2026*, AIAA, Jan. 2026. doi: 10.2514/6.2026-2022.

[47] S. G. Manyam, D. W. Casbeer, **A. Von Moll**, and I. Weintraub, “Dubins Path with Terminal Range and Field-of-View Constraints,” in *Conference on Decision and Control*, IEEE, Dec. 2025. doi: 10.1109/LCSYS.2025.3582622.

[48] P. Surve, R. L. Frost, S. D. Bopardikar, **A. Von Moll**, and D. W. Casbeer, “Heterogeneous Pursuit of Multiple Translating Targets,” in *Conference on Control Technology and Applications*, IEEE, Aug. 2025. doi: 10.1109/CCTA53793.2025.11151369.

[49] A. W. Denton, D. Kunz, I. E. Weintraub, and **A. Von Moll**, “The Basic Engagement Zone in 3-D,” in *National Aerospace Electronics Conference*, IEEE, July 2025. doi: 10.1109/NAECON65708.2025.11235322.

[50] I. Weintraub *et al.*, “Min-Time Escape of a Dubins Car from a Polygon,” in *American Control Conference*, IFAC, July 2025. doi: 10.23919/ACC63710.2025.11107564.

[51] S. G. Manyam, D. W. Casbeer, **A. Von Moll**, and I. Weintraub, “Shortest Dubins Path to a Moving Circle with Free Final Heading,” in *American Control Conference*, IFAC, July 2025. doi: 10.23919/ACC63710.2025.11107508.

[10] **A. Von Moll**, D. Milutinović, I. Weintraub, and D. W. Casbeer, "One-vs-one Threat-Aware Weaponeering with Basic Engagement Zones," in *International Conference on Unmanned Aerial Systems*, IFAC, May 2025, pp. 1138–1145. doi: 10.1109/ICUAS65942.2025.11007913.

[52] T. Chapman, I. Weintraub, **A. Von Moll**, and E. Garcia, "Engagement Zones for a Turn Constrained Pursuer," in *International Conference on Unmanned Aerial Systems*, IFAC, May 2025, pp. 763–768. doi: 10.1109/ICUAS65942.2025.11007894.

[53] A. Wolek, D. Casbeer, I. Weintraub, and **A. Von Moll**, "Maximum Kinetic Energy Paths for a Dubins Vehicle with Decaying Speed," in *SciTech*, AIAA, Jan. 2025. doi: 10.2514/6.2025-1351.

[54] I. Weintraub, **A. Von Moll**, A. Hansen, and M. Zollars, "An Optimal Strategy for Off-Board Proximal Sensing of a Target: Part 1," in *SciTech*, AIAA, Jan. 2025. doi: 10.2514/6.2025-1349.

[55] A. Hansen, M. Zollars, I. Weintraub, and **A. Von Moll**, "An Optimal Strategy for Off-Board Proximal Sensing of a Target: Part 2," in *SciTech*, AIAA, Jan. 2025. doi: 10.2514/6.2025-1350.

[56] D. Milutinović, **A. Von Moll**, I. E. Weintraub, and D. Casbeer, "Policy for Optimal Avoidance of Multiple Engagement Zones," in *SciTech*, AIAA, Jan. 2025. doi: 10.2514/6.2025-1544.

[57] B. Mora, A. Chakravarthy, **A. Von Moll**, I. E. Weintraub, and D. Casbeer, "Influence of Malicious Agents in a Team Seeking to Contain an Evader," in *SciTech*, AIAA, Jan. 2025. doi: 10.2514/6.2025-1348.

[58] D. Milutinović, **A. Von Moll**, S. G. Manyam, D. W. Casbeer, I. E. Weintraub, and M. Pacther, "Pursuit-Evasion on a Sphere and When it Can Be Considered Flat," in *Conference on Decision and Control*, IEEE, Dec. 2024. doi: 10.1109/CDC56724.2024.10885887.

[59] A. Wolek, I. E. Weintraub, **A. Von Moll**, D. Casbeer, and S. G. Manyam, "Sampling-Based Risk-Aware Path Planning Around Dynamic Engagement Zones," in *Modeling, Estimation and Control Conference*, IFAC, Oct. 2024. doi: 10.1016/j.ifacol.2025.01.030.

[60] **A. Von Moll**, A. R. Gerlach, C. Bakker, A. Rupe, and M. Pachter, "Constrained Turret Defense with Fixed Final Time," in *Modeling, Estimation and Control Conference*, IFAC, Oct. 2024. doi: 10.1016/j.ifacol.2025.01.121.

[61] B. Mora, **A. Von Moll**, I. Weintraub, D. W. Casbeer, and A. Chakravarthy, "Escape from an Orbiting Pursuer with a Nonzero Capture Radius," in *National Aerospace Electronics Conference*, July 2024.

[62] D. Maity, **A. Von Moll**, D. Shishika, and M. Dorothy, "Optimal Evasion From a Sensing Limited Pursuer," in *American Control Conference*, July 2024. doi: 10.23919/ACC60939.2024.10644990.

[63] I. E. Weintraub, **A. Von Moll**, and M. Pachter, "Minimum Time Escape from a Circular Region of a Dubins Car," in *National Aerospace Electronics Conference*, IEEE, July 2024. [Online]. Available: https://arxiv.org/abs/2405.05725

[64] I. E. Weintraub, **A. Von Moll**, D. W. Casbeer, and S. G. Manyam, "Virtual Target Selection for a Multiple-Pursuer Multiple-Evader Scenario," Jan. 2024. doi: 10.2514/6.2024-0123.

[65] **A. Von Moll**, D. W. Casbeer, I. E. Weintraub, and M. Pachter, "Pure Pursuit of a Target on a Circular Trajectory," in *AIAA SciTech*, Jan. 2024. doi: 10.2514/6.2024-0956.

[66] I. Weintraub, **A. Von Moll**, and M. Pachter, "Range-Limited Pursuit-Evasion," in *2023 National Aerospace and Electronics Conference (NAECON)*, Aug. 2023. doi: 10.1109/NAECON58068.2023.10365808.

[67] S. Bajaj, S. Bopardikar, **A. Von Moll**, E. Torng, and D. Casbeer, "Perimeter Defense using a Turret with Finite Range and Startup Times," in *American Control Conference*, IEEE, May 2023. doi: 10.23919/ACC55779.2023.10155838.

[68] A. Pourghorban, M. Dorothy, D. Shishika, **A. Von Moll**, and D. Maity, "Target Defense against Sequentially Arriving Attackers," in *Conference on Decision and Control*, IEEE, Dec. 2022. doi: 10.1109/CDC51059.2022.9992425.

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[71] Z. E. Fuchs, **A. Von Moll**, and D. Casbeer, "Engage or Retreat Differential Game with N-Targets and Distributed Defensive Assets," in *Conference on Controls Technology and Applications*, Aug. 2021. doi: 10.1109/CCTA48906.2021.9658922.

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[79] E. Garcia, **A. Von Moll**, D. Casbeer, and M. Pachter, "Strategies for Defending a Coastline Against Multiple Attackers," in *IEEE Conference on Decision and Control*, Dec. 2019. doi: 10.1109/CDC40024.2019.9029340.

[80] S. G. Manyam, D. Casbeer, **A. Von Moll**, and Z. Fuchs, "Optimal Dubins Paths to Intercept a Moving Target on a Circle," in *American Control Conference*, July 2019. doi: 10.23919/ACC.2019.8814913.

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